Commodity markets and price formation: a working primer

Compiled November 8, 2026 by Claude Code from six sourced research notes in research-notes/. This is background for the NG discovery program: basic understandings to build from and argue with, not rules. Where the literature is settled it says so; where it is contested or unknown it names the competing explanations and the observation that would separate them. It contains none of the project's own October findings.

Sourcing caveat: primary hosts (cmegroup.com, eia.gov, cftc.gov, arxiv.org, nber.org) were outside this session's network allowlist, so numbers come from search summaries, abstracts and secondary pages. Items marked (secondary) or (verify) should be confirmed against the linked primary page before being quoted as hard facts. Each note file lists its own gaps.

1. What a futures contract is and how Henry Hub natural gas trades

A futures contract is a cleared obligation, marked to market daily against the exchange settlement price, with initial and maintenance margin posted at the clearing house. A margin call restores equity to the initial level, not the maintenance level, which makes forced liquidation lumpier than in securities margin accounts. CME margin education

NYMEX Henry Hub Natural Gas (NG) specifics:

Item	Value	Source
Contract size	10,000 MMBtu	CME specs
Tick	$0.001 per MMBtu, worth $10 per contract	CME specs (secondary confirmation)
Futures expiry	Third-last business day of the month before the contract month	CME specs
Standard options expiry	Fourth-last business day, one business day before the futures	CME options guide
Delivery	Physical, at Sabine Pipe Line's Henry Hub, Erath, Louisiana, ratably over the month	CME specs; NYMEX Chapter 220 (not retrieved)
Globex hours	About 6:00 pm to 5:00 pm ET, Sunday to Friday, 60-minute daily break (secondary)	StoneX
Regular trading hours	Begin 9:00 am ET	Kurov et al. summary

Item	Value	Source
Daily settlement	VWAP of outright trades in the 14:28 to 14:30 ET window, rounded to the tick; deferred months from spread VWAPs anchored to the front, 100-contract minimum; fallback to last trade checked against the 14:30 bid and ask	NYMEX filing 2019, 2015
Circuit breakers	Dynamic, rolling 60-minute window; the level (10 percent per the current page, 15 percent per a March 2020 notice) in force in October 2023 is unresolved (verify)	CME, SER-8569

Three things follow for intraday work. The settlement is a two-minute VWAP, so anyone benchmarked to it must trade through the window, which concentrates volume and makes the window's price a liquidity-weighted average rather than a marginal price; the curve (deferred months) is fixed in the same two minutes through spread prints. Options expire the day before the futures, so the penultimate day of the front month carries exercise and assignment flows plus final futures positioning. The settlement window under EDT (through November 4, 2023) is 18:28 to 18:30 UTC; after the November 5 clock change it is 19:28 to 19:30 UTC (inference; verify).

Continuous series: a single contract month needs no adjustment; splicing months requires a declared method (difference or ratio back-adjustment), and each method breaks a different property (point value or percentage change). Quantpedia

2. Who trades and why: the two theories of the curve

The CFTC classifies reporting traders by predominant business purpose into Producer, Merchant, Processor or User; Swap Dealer; Managed Money; and Other Reportables, as of each Tuesday's open interest. Staff classify traders, not trades. CFTC explanatory notes

Two frameworks explain risk premia and the shape of the curve, and they are not settled against each other:

- **Hedging pressure (Keynes, Hicks; Bessembinder 1992; de Roon, Nijman and Veld 2000).** Net hedger positioning earns speculators a premium; own and cross hedging pressure explain futures returns across 20 markets. de Roon et al.. Kolb (1992) found "normal backwardation is not normal" across 29 contracts (secondary).
- **Theory of storage (Kaldor, Working; Fama and French 1987; Gorton, Hayashi and Rouwenhorst 2013).** Basis = interest plus storage cost minus convenience yield, with inventories as the state variable. Convenience yield is a decreasing nonlinear function of inventories; low inventories go with backwardation, high with contango; GHR find no evidence that participants' positions predict premia. NBER w13249

Kang, Rouwenhorst and Tang (2020) partly reconcile them: hedging pressure mixes speculators' short-horizon liquidity demand and hedgers' long-horizon insurance demand, which move expected returns with opposite signs. JF 2020

For natural gas the seasonal cycle is storage theory: injection season (nominally April to October, not strictly) builds inventory; the winter strip carries a scarcity premium whose size depends on how full storage is. The March/April spread (the "widowmaker") is the market's price for end-of-winter risk; it

reached a record $2.13 in May 2022 and a record-low premium in 2023. EIA on seasons, Oil and Gas 360

3. Natural gas fundamentals and their clocks

The storage report. The EIA Weekly Natural Gas Storage Report is released Thursday at 10:30 am ET (14:30 UTC), covering working gas as of 9:00 am the prior Friday, so the number is six days stale at release. Holiday weeks move it to Wednesday noon or Friday. It is sampled through Form EIA-912 from the EIA-191 census and is EIA's only Principal Federal Economic Indicator. EIA methodology, schedule

Consensus is accurate: the median Bloomberg forecast anticipates over 99 percent of the variation in storage changes and is impounded before release; analyst error averaged 13.2 Bcf in a 1997 to 2004 sample, larger in withdrawal season. Ederington et al. 2019, Gay, Simkins and Turac via NGI

What the report does to prices, from the announcement literature:

Finding	Source
Volatility elevated at release and for about 30 minutes after; also heavy at open and close	Linn and Zhu 2004
Price moves inversely to the storage surprise, on release day only; futures lead spot; response roughly doubled after 2005	Chiou-Wei, Linn and Zhu 2014
OLS understates the impact because the surprise variable is noisy; corrected estimates about four times larger for NG	Halova, Kurov and Kucher 2014
Pre-announcement drift begins about 90 minutes before, at the 9:00 am open, and is correlated with the surprise	Kurov et al. 2018
More than half of NG futures' annual return accrues on announcement days; the pre-announcement return occurs only ahead of bearish surprises, which argues against simple informed trading	Prokopczuk, Wese Simen and Wichmann 2021

The coefficient in cents per Bcf was not retrievable from any of these.

Weather. GFS and ECMWF run at 00, 06, 12 and 18 UTC; ECMWF products arrive about 08:30 UTC (00z) and 20:30 UTC (12z), provisional. The midday (12z) run is the one traders watch; a 2013 source puts the market-moving window at 11:00 am to 1:00 pm ET. Weather affects about half of US gas demand; weather surprises move both the mean and the volatility of NG returns. NOAA GFS, ECMWF schedule, Mu 2007

A lesson from orange juice: Roll (1984) found weather explained little of OJ futures variance; Boudoukh et al. later showed the puzzle dissolves once you condition on the nonlinear freeze threshold. The same logic applies to heating demand: condition on the HDD regime before concluding weather does not matter. Boudoukh et al.

Why gas is so volatile. Both supply and demand are inelastic in the short run, storage is finite, and the system is pipeline-bound and regional, so any fundamental shock produces large swings. Crude volatility predicts gas volatility but not the other way round; volatility shocks are short-lived. Front-

month annualised volatility was about 81 percent in Q4 2024 and about 69 percent by mid-2025 (secondary), so thresholds in absolute dollars are non-stationary by construction. EIA 2003, Pindyck 2003

October 2023 context. Prompt below $2.90 early in the month, spot above $3 for the first time since January in the week of October 5, $3.18 on October 11 and $2.90 on October 18; storage 3,700 Bcf on October 20, 182 Bcf (5.2 percent) above the five-year average; record production (113.1 Bcf/d marketed), record LNG feedgas, record consumption. A comfortable-storage, record-supply regime with a compressed winter premium. EIA NGWU Oct 19, AGA Oct 31

4. What a price movement is, mechanically

In a limit order book the midpoint changes in exactly two ways: the best queue on one side reaches zero (by fills or by cancellations), or a new order is posted inside the spread. Trade size is a weak cause; the depth sitting at the best price is the strong one. Large price changes come from gaps in the book, not from large trades. Farmer et al. 2004

On CME market-by-order data (GLBX.MDP3 via Databento) every add, modify, cancel, fill and trade summary is visible with order identity and FIFO priority. The book is valid only on records flagged `F_LAST` (end of a matching event per instrument); between those records the displayed best may already have traded. `ts_event` is the matching engine clock, `ts_recv` the capture clock. Databento has announced that from July 2026 the `F_LAST` flag moves to a separate `action='N'` record. Iceberg refreshes keep the same order ID and go to the back of the queue, so hidden liquidity on Globex is detectable after the fact as repeated same-ID refreshes at one price. Databento GLBX.MDP3, Databento 2026 change, CME matching steps

A sweep is one matching event: a trade summary (T) with the aggressor side, fills (F) against resting orders at successively worse prices, then cancels or modifies for the orders that were exhausted. A trade moves the quote only if the aggressor's size at least equals the displayed size of the queue it hits. On CME the aggressor side is supplied by the exchange, so trade signing is exact; the Lee-Ready problem (75 to 85 percent accuracy in equities) does not arise for trades, only for the question of which non-trade events to count as flow. Ellis, Michaely and O'Hara

Classic theory and what it predicts about the price response to a trade:

- **Kyle (1985):** linear, permanent impact; lambda is the inverse of depth.
- **Glosten and Milgrom (1985):** the spread is the cost of adverse selection; each trade moves the quote permanently by the market maker's Bayesian update.
- **Inventory models (Ho and Stoll; Amihud and Mendelson):** transient pressure that reverts as the dealer rebalances, giving negative autocorrelation in price changes.
- **Hasbrouck (1991):** the cumulative impulse response of the quote to a trade innovation separates the permanent (information) plateau from the transient (inventory) overshoot.
- **Roll (1984):** bid-ask bounce alone creates negative autocorrelation in trade prices; it is measurement, not a return.

Empirical regularities that replicate:

Regularity	Scope of evidence	Source
Short-horizon midpoint change is linear in order-flow imbalance at the best quotes, slope about 1/depth; adds and cancels carry impact comparable to trades, and dominate in large-tick books	50 NYSE stocks; US equities	Cont, Kukanov and Stoikov, Eisler et al.
Queue imbalance (bid size minus ask size over their sum) predicts the next midpoint move strongly in large-tick instruments (spread pinned at one tick) and weakly in small-tick ones	Nasdaq LOBSTER	Gould and Bonart
Metaorder impact grows as about the square root of size (exponent 0.5 to 0.6), measured on futures	CFM futures metaorders 2007 to 2010, about 500,000 trades	Toth et al. 2011
Order flow has long memory (Hurst about 0.6 to 0.8) while prices are nearly uncorrelated; the persistence is order splitting by single institutions, not herding	LSE with member IDs	Lillo and Farmer, Toth et al. 2015
Impact of a metaorder decays after it ends, to roughly two-thirds of peak within the day (contested whether to zero)	ANcerno US equities	Bucci et al., Brokmann et al.
Event arrivals are self-exciting; the Hawkes branching ratio is near one in E-mini futures, though the "rising reflexivity" reading is kernel-dependent	E-mini S&P 1998 to 2012	Hardiman, Bercot and Bouchaud, Filimonov and Sornette
After an aggressive order, spreads and depth recover within about 20 best-limit updates, with partial price reversal; fast replenishment happens only in a minority of cases	Paris Bourse; LSE	Degryse et al., Large 2007 (not fetched)
Liquidity vanishes when providers with no quoting obligation cancel ahead of being hit; the 2010 flash crash was a large selling program meeting a withdrawing book; HFTs did not trigger it but exacerbated volatility	CFTC audit trail, E-mini	Kirilenko et al.

The single most important moderator is tick size. In a one-tick-spread book the only way the midpoint moves is a queue depleting, so the ratio of queue sizes is the race, and imbalance predictability is strong; in multi-tick books it collapses. Whether front-month NG at $0.001 is a large-tick book in the day session is an empirical question the tape can answer; back months and overnight will be small-tick. Results should never be pooled across the two regimes. Dayri and Rosenbaum

Exchange mechanisms interrupt normal dynamics during fast moves: CME Velocity Logic (reserves the market when price moves too far too fast within a one-second window) and Stop Logic (reserved state when triggered stops would move price beyond the no-bust range; it produced the five-second pause on May 6, 2010). Reserved states accumulate order flow without price response and should be excluded like the settlement window. CME Velocity Logic

Peer-reviewed microstructure work on NYMEX natural gas books specifically is thin to absent; depth, spread-in-ticks, resiliency and settlement-period behaviour have to be measured from the project's own data.

5. Market psychology: what is established and what is not

Established at the individual level, almost entirely in equities and retail accounts at daily to monthly horizons: the disposition effect (selling winners, holding losers), underperformance of high-turnover retail traders (Taiwan day traders: 97 percent expected to lose, under 1 percent persistent winners), attention-driven buying, and extrapolative survey expectations that are negatively correlated with subsequent returns. Barber, Lee, Liu, Odean, Zhang, Greenwood and Shleifer

Most damaged: loss aversion as a general, stake-independent law (Gal and Rucker 2018, rebutted by Mrkva et al. 2020). Fama's 1998 objection stands: overreaction and underreaction appear about equally often and anomalies are methodology-sensitive. Lo's adaptive markets hypothesis reframes biases as heuristics that work in some regimes and not others, which maps onto per-cell evaluation. Gal and Rucker, Lo 2004

The futures evidence is thin and old but direct:

- Coval and Shumway (2005), CBOT T-bond locals, 1998 audit trail: traders with morning losses were about 16 percent more likely to take above-average afternoon risk; they bought above and sold below prevailing prices; their price impact reverted faster than others' and other traders faded them. A behavioral footprint exists on the tape and is transient. CFA Digest
- Locke and Mann (2005), CME floor locals: losses held longer than gains, framed as "discipline"; exact figures not retrieved.
- Boyd et al. (2016), 30 US futures markets 2004 to 2009: managed-money herding is real, does not destabilize prices, and declines with electronic trading. JFM 2016
- Osler (2003), one FX bank's orders: take-profits cluster at round numbers, stops just beyond them, so support and resistance bounces and post-break accelerations are mechanically self-fulfilling; stop cascades produce continuation, not the retail "sweep and reverse". No comparable study exists for CME energy. NY Fed SR 125, SR 150
- Menkhoff (2010), 692 fund managers: technical analysis dominates at weekly horizons, and its users believe prices are psychologically driven. JBF 2010

The central methodological point for tape work: every behavioral footprint has an observationally equivalent mechanical generator.

Behavioral story	Mechanical twin	What separates them
Herding, copying	Order splitting of one institution's parent order; common reaction to a public number	Participant IDs; timing relative to the release
Panic, capitulation	Stop orders converting to market orders; margin liquidation; market makers re-centering quotes	Order-type data; whether cancelled IDs re-appear at new prices
Loss-chasing	Inventory-constrained market makers forced to unwind	Account P&L (audit trail only)
Anchoring to round numbers	Option strikes and dealer gamma hedging; algorithms quoting one tick ahead of the cluster	Strike open interest; order placement relative to the cluster
Attention-driven buying	The news itself; hedging flows it triggers	Does the drift revert (attention) or persist (information)

Attribution to psychology has only ever been achieved with participant-identified data. On an anonymous tape the behavioral layer is a set of named alternatives whose predictions about timing and reversal can be tested per cell; it is not a source of direction on its own. Physiological studies (cortisol tracks uncertainty, testosterone tracks P&L, 10 to 17 traders) establish a channel but no price footprint. Coates and Herbert

6. What quants look for and how they avoid fooling themselves

Quants hunt four kinds of edge: risk premia, behavioral anomalies, liquidity provision and microstructure, and information speed. The published record says most discoveries are inflated: anomaly returns fall about 26 percent out of sample and about 58 percent after publication; a published strategy's Sharpe roughly halves; a new factor should clear t greater than 3.0, not 2.0, given cumulative data mining. McLean and Pontiff, Harvey, Liu and Zhu, CFM half-life study

Commodity signals that have survived decades of replication are all slow: time-series momentum (positive in every one of 58 contracts 1985 to 2009; positive every decade since 1880 across 67 markets), carry and term structure (within-class Sharpe 0.5 to 0.9), hedging pressure sorts (contested by GHR), basis-momentum (JF 2019). None speaks to minute-scale behavior. The intraday evidence that exists for energy is event-time structure: the storage release and open and close volatility. Moskowitz, Ooi and Pedersen, Koijen et al., Boons and Prado

The validation protocol, and why it matters with few independent days:

- The unit of independence is the day or the event, not the tick. CFM's identity t = Sharpe times the square root of years says no tick-level Sharpe is significant on its own; about 50 storage releases per year per contract is the true sample size for any announcement-conditioned claim. CFM Lucking Up
- Purge training samples whose label window touches the test period and embargo the following session; bootstrap at the block or day level (stationary bootstrap, block length at least a session). Overlapping forward windows share increments, so the effective sample is about N/k and naive standard errors are overstated by about the square root of k. Politis and Romano, purged CV
- Track the number of trials. Every cell, threshold and variant examined is a test; deflate for them (deflated Sharpe ratio, probability of backtest overfitting, minimum backtest length about 2 ln(N) over the squared expected maximum). Bailey and Lopez de Prado
- Sullivan, Timmermann and White: the best of a large universe of technical rules shows no genuine outperformance once data snooping across the universe is accounted for. JF 1999
- Score probabilistic forecasts with a proper rule (Brier) and its reliability component, not hit rates; a three-way up, flat, down target keeps flats in the denominator.
- Look-ahead through the signal definition is the subtle one: a feature computed over an interval whose end is defined by the outcome (the leg ends when price reverses; the metaorder ends when the trader stops) is conditioned on the future.

Machine learning: Gu, Kelly and Xiu report monthly out-of-sample R squared of about 0.3 to 0.4 percent and a 1.35 Sharpe long-short, which a replication thesis could not reproduce. Order-book deep learning (Sirignano and Cont; DeepLOB) reaches about 70 percent next-move accuracy on

equities, but the 2023 cross-dataset benchmark found most models overfit the FI-2010 set and degrade sharply on unseen data. Microstructure predictability is real at the next-event horizon and is venue-, tick- and regime-specific. Sirignano and Cont, Prata et al.

Practitioner habits that recur: Renaissance's three steps (find the anomaly, confirm it is significant and consistent, then seek an explanation but keep it if none is found); most academic strategies failed in-house replication; Carver's "weight, do not drop" rule for weak signals and small simple rule pools; CFM's point that portfolio Sharpe grows with the square root of the number of independent signals, so many weak signals beat one strong one. Impact is the bridge from gross signal to tradeable edge: the square-root law means edge over impact deteriorates as the square root of size, which defines capacity. The project excludes trade cost from research, but the concept is why a small hit-rate edge can still exist (Medallion's reported 50.75 percent, second-hand). Zuckerman notes, Carver via CXO

7. What nobody has explained

Regularities everyone accepts whose causes are open:

Puzzle	Status	Competing explanations	What would separate them
Fat tails (return tail exponent about 3)	Regularity settled, mechanism open	Large-trader size plus concave impact (Gabaix et al.) vs endogenous self-excitation and liquidity criticality; Farmer and Lillo show volume alone fails	Condition tails on volume and trade frequency across markets; test whether exponents move with institutional concentration
Volatility clustering and long memory	Regularity settled; "rough volatility" is contested as a fact (Cont and Das: realized volatility looks rough even when spot volatility is not)	News-arrival mixture vs interacting heterogeneous agents vs order splitting	Compare clustering persistence on release vs non-release days; apply the roughness estimator where microstructure noise is absent
Most large moves have no news	Replicated at every resolution (Cutler, Poterba and Summers 1989; Fair 2002; Joulin et al. 2008; Ait-Sahalia and Li 2024, who also rule out insider trading as the bulk explanation)	Flow and inelastic demand vs endogenous liquidity dynamics vs discount-rate variation	Post-jump volatility relaxes differently after news jumps and liquidity jumps; exogenous flow shocks with no information content give the multiplier
Long-memory order flow with martingale prices	Settled; splitting origin is near-consensus; the compensating mechanism is open	Power-law decaying impact propagator (Bouchaud et al.) vs asymmetric liquidity (Lillo and Farmer) vs multi-timescale latent book	Impact trajectories when a metaorder is interrupted or reversed
Square-root impact	Broadly accepted; exact exponent and universality contested	Latent liquidity vanishing at the price (Toth, Donier) vs Kyle-Obizhaeva invariance	Exponent as a function of participation rate; bet-size scaling across assets

Puzzle	Status	Competing explanations	What would separate them
Momentum and reversal coexisting	Trend profitable over a century; cause open	Underreaction then overreaction vs hedging premia vs slow institutional flows; reversal as compensation for liquidity provision (Nagel)	Trend returns should scale with asset-level flow persistence if flow-driven; momentum crashes are partly forecastable, which a pure risk story cannot absorb
Inelastic markets ($1 of flow moves aggregate value about $5)	Open; identification under attack	Mandate-constrained institutions vs persistence contaminating elasticity estimates (van Binsbergen, David and Opp)	Elasticity estimates should fall as the horizon passes the instrument's persistence; roll-window flows in futures
Commodity index "financialization"	Masters bubble hypothesis mostly rejected; changed co-movement mostly accepted	Irwin and Sanders vs Singleton (significant flow effects on imputed positions) and Tang and Xiong	The index versus non-index co-movement gap should have narrowed after index AUM fell

Commodity-specific puzzles:

- **Natural gas announcement-day puzzle.** More than half the annual return on EIA days, with the pre-announcement return realized only ahead of bearish surprises; practitioners record identical surprise signs producing opposite moves within a week. No systematic test of post-announcement asymmetry was retrievable. Prokopczuk et al.
- **Futures as forecasts.** Energy futures are roughly unbiased but rarely beat a no-change forecast; EIA forecasts are worse than no-change; the one robust conditional is that futures beat the random walk when the basis exceeds about 5 percent. Alquist, Kilian and Vigfusson, FEDS Notes 2026
- **Negative WTI, April 20, 2020.** The CFTC interim report named fundamentals plus unspecified "technical factors" and, by a commissioner's own statement, did not identify a root cause; academics converge on expiry and trade-at-settlement closing pressure. CFTC release, Ge et al. AEA 2023
- **Samuelson effect** (volatility rising toward expiry) is absent in most contracts and present in about 30 percent of energy contracts; where it holds, the Bessembinder covariance mechanism fits. Daal, Farhat and Wei
- **Amaranth and the widowmaker.** The loss ($4.9 to $6.4 billion depending on source) is documented; a statistical study found no significant effect of its positions on prices or spreads, while contemporaries said it propped up the curve. Contested.
- **Absorption and exhaustion** as practitioners define them have no academic test; the nearest published relatives are iceberg detection and replenishment on CME (Zotikov and Antonov 2021), detected icebergs attracting market orders (Frey and Sandas), and a crypto preprint finding that reversal after flow-driven moves scales with the flow and not with any hole in the book. Zotikov and Antonov, arXiv 2608.21888

8. Unconventional framings and what they have actually earned

Ranked by evidence, from the novel-frameworks note:

Strong evidence. The square-root impact law on futures (Toth et al.). Intraday momentum from option-dealer gamma hedging across more than 60 futures including commodities, strongest when dealers are short gamma (Baltussen, Da, Lammers and Martens, JFE 2021). Order flow explaining more than half of daily FX changes and beating a random walk out of sample (Evans and Lyons). Broker-dealer risk capacity forecasting energy futures returns in and out of sample (Etula 2013). Zero-intelligence order flow explaining 96 percent of cross-sectional spread variance and 76 percent of volatility variance (Farmer, Patelli and Zovko), with the caveat that it gets direction right and magnitude wrong. Latency-arbitrage races about once per minute per symbol (Aquilina, Budish and O'Neill). Baltussen et al., Evans and Lyons, Etula, Farmer et al.

Useful frames with indirect support. The latent order book (Donier et al.): the visible book is mostly market makers; the real supply and demand is a hidden reservoir of intentions, vanishingly thin near the price, which makes small orders disproportionately impactful and predicts partial reversion when a flow stops. Market ecology (Farmer; Scholl, Calinescu and Farmer): strategies are species competing for the same flow, returns are density-dependent, and edges are consumed as capital crowds in; demonstrated in a toy model, not measured on real strategy capital. Kyle-Obizhaeva invariance: bet size, arrival rate and impact scale with dollar volume and volatility in "business time", which gives a dimensional null for how much a book should move per contract. Donier et al., Scholl et al., Kyle and Obizhaeva

Contested. Log-periodic bubble prediction (Sornette): some ex-ante calls, documented misses, and the functional form is "sloppy" under small sample changes. The Hawkes branching ratio read as rising "reflexivity": kernel-choice dependent, with regime switching producing spurious criticality. Self-organized criticality inferred from power laws: log-normals fit as well (Clauset, Shalizi and Newman). Risk-parity feedback: accepted sign, disputed size.

Narrative only. Soros's reflexivity at any intraday level; Minsky at trading frequency; Market Profile value area and the "80 percent rule"; Wyckoff; footprint delta divergence; "stop hunt then reverse" (the evidence points to continuation after a stop cluster trades); "markets as measurement devices" and order-book conservation laws beyond the reaction-diffusion annihilation picture.

Two framings worth carrying as hypotheses because they generate testable order-book predictions: flow and inventory pricing (the sign of recent aggressive flow should explain the next move far better than any fundamental at seconds horizons, and impact per unit flow should rise when intermediary capital is constrained), and the latent-book conservation picture (a move driven by a transient-inventory holder must be unwound and so reverts; a move driven by a hedger changing open interest need not).

9. What an order-book researcher can and cannot see

Observable on CME MBO: every displayed order's birth, size changes, price changes, fills and removal, with queue priority; the aggressor side of every trade; iceberg refreshes by order ID; spread, depth at every level, and their changes event by event; the matching engine clock and the capture clock; exchange reserved states; declared publication clocks for scheduled releases.

Not observable: who owns an order; whether two orders belong to one participant (order splitting versus herding); the undisplayed remainder of an iceberg before it refreshes; stop orders before they trigger; positions, P&L, margin state and risk limits; whether a cancellation was fear, repricing or a hedge completing; beliefs, intent and hormones. Implied liquidity from calendar spreads appears in aggregated depth but not as individual orders (unconfirmed for GLBX.MDP3).

Consequences: trade signing is exact but flow definition is a choice; book state is valid only at event boundaries; a quote can be stale by seconds in a quiet book, so every "pre-event" snapshot carries an age; every behavioral reading has a mechanical twin that the anonymous tape cannot separate without a timing or reversal test; the day or the release, not the tick, is the unit of independence; and fundamentals enter the intraday tape mostly at fixed clocks (10:30 ET Thursdays, model runs at 00, 06, 12, 18 UTC, the 14:28 to 14:30 ET settlement), with the rest of the tape's movement attributable, on present evidence, mostly to the flow and liquidity dynamics in sections 4 and 7.

Appendix A. Project carry-over: the dipole applications and the NG look-ahead time budget

This appendix is project knowledge from earlier sessions (S36, S80 to S92), not literature. It is kept here so the two meanings of "dipole" and the measured clocks travel with the primer.

Two different objects are both called "the dipole".

Name	What it measures	Window and clock	Status
`dip_imb_level` (S92 direction nowcast)	Lee-Ready buy-minus-sell flow imbalance around a leg's trigger push	about 300 s, overlapping the leg's birth	Direction only: low tercile 7 percent u monotone with strength (0.68 at unde 0.30), out of sample 34 of 34 on stron unseen warm-season days. A nowcast correctly signed legs still round-trip. `research/kalshi/NG_BEHAV` `knowledge/ng_brain.json` `direction.flow_nowcast`
`imb_R` (resting-book contrarian tilt)	Resting-book imbalance at entry	at entry	Ask-heavy book went up 0.61 (n 158 121); when flow and book disagree fl breaker only
Divergence and exhaustion filter (S36, `odcore/info_dipole.py`)	`aligned_flow = imb_level x sign(price_drift)`; exhaustion = the imbalance collapsing toward 0.5	crypto era, 1 s bins	Crypto: divergence at or below minus percent reversal; oppose plus exhaust plus strengthen 49 percent. NG canar trades): static divergence did not trans below aligned); exhaustion showed a W60/K30/F60 (0.410 vs 0.382). S92 `dip_aligned_flow`, `dip_mi_` were noise for direction, sustain, turn season days

Name	What it measures	Window and clock	Status
Archived Native candidate detector (the October 18 "dipole" in the discovery logs)	Peak of the 20-second rolling signed-lot ratio above a trailing 85th-percentile bar, SAME or FLIP against the predecessor	recognized at a later completed group; median 2.407 s after the nominal second, maximum 87.4 s	A different object from `dip_imb_l` about it on October 18 bears on the S result does not validate it

Also carried over: the turning-point read (held big legs recruit far-side liquidity, `turn_far_thinning` median minus 0.156 versus 0.000 for reversed tops, n 101), the magnitude staircase (continuation 0.92 once a leg crosses $350, 11 of 11 at $500), and the leakage gate `odcore/leakage.py` before any backtest. The crypto-era fee-floor rules and the `DEPLOY` cell map are retired and do not apply to NG.

The NG look-ahead time budget (the clocks the program has measured).

Clock	Measured value	Source and scope
Futures lead Kalshi, 1-minute bars	WTI: 41 contracts, 15 significant, peak lags {0 min: 6, 1 min: 8, 5 min: 1}; futures lead or synchronous, Kalshi never leads. Gold and silver: same shape, 0 to 1 minute. NG at 1 minute not measured (Pyth has no gas feed; Yahoo 1-minute only)	S80, S91; `futures_kalshi_lag.py`
Futures lead Kalshi, sub-minute	On a 17 cent WTI jump the most liquid strike's first reprice tick came about 7 s later and the reprice continued for about 20 s; less liquid strikes lag longer. One event; the NG-specific sub-minute lag is still unmeasured	S81
NG release move speed	60 s after the 14:30 UTC storage print captures a median 0.66 of the full move; 60 s move p50 106 bps (310*percontract*), *p*90174*bps*(490); one day peaked in 1 s, five of twelve within 2 minutes; the pooled median time to peak (about 12 min) hides this	S85, `EVENT_MOVE_FINDINGS_S85.md`, 12 releases Apr to Jul 2026
NG lag-join echo	Release mode NG taker plus 17 cents, 100 percent, tiny n; provisional	S87, `lag_join.py`
Direction nowcast window	about 300 s of flow overlapping the leg's trigger push	S92
Candidate recognition delay	median 2.407 s after the nominal available second, maximum 87.4 s; the available second is itself t0+50 (5 s local maximum plus 45 s refractory), so actual recognition is a median 51 s after the burst (range 7 to 137 s)	this session's reconstruction, October 18

Clock	Measured value	Source and scope
Exhaustion runway under the frozen contract	structural onset a median 31 to 34 s after t0 on October 2, 17 and 18; persistent (P) family 110 to 121 s, collapsed families 20 to 24 s	this session's whole-day pass, three dates
State-confirmation versus information horizon	62 to 63 percent of exhaustions are confirmed over before t0+50; 68 to 69 percent before the seed state is knowable at t0+61; the 85th-percentile threshold saturates at 1.0 on all three days	this session's whole-day pass, three dates
Next print after a decision close	median lead 0.366 ms; 63 of 29,797 had a zero receive-time lead with tape order still separating them	October 18 prebirth log
Paper replay entry assumption	first completed book at least 1 ms after the signal group	historical paper replay
Settlement reference	KXNATGASD settles on the Pyth NGDQ6 1-minute candle close at 5:00 pm EDT (21:00 UTC), not the CME settle; the project guards 1,800 s before it	S84, S87
Sampling floor	1 s historical sampling undersamples NYMEX and is a lower bound; MBO is the actual tape	S84

Read together: if the exploitable Kalshi window is on the order of 7 to 20 s after a NYMEX move, a detector whose recognition arrives a median 51 s after the burst, and after the burst itself has ended in five of eight cases, cannot serve that window at all; its post-endpoint plays start at confirmation plus 5 s by design. The NG release move itself is two-thirds done within 60 s and sometimes within 1 s. Any signal meant for the lag has to be judged on its recognition clock, not its nominal clock, against these numbers.

Claude Code, November 8, 2026. Sourced notes: tasks/family-research-20261022/claude-code/research-notes/ on branch codex/ng-shadow-discovery-20261108. Items marked (secondary) or (verify) were not read from the primary page in this session. Appendix A is project carry-over, not literature.